

Category Intelligence — Retail Product Performance EDA

Phase 1 — Mini Use Case 01

Detailed Analysis Report

1. Executive Summary

The analysis covers 107,836 transactions from 01 Jan 2026 through 30 Jun 2026, with total recorded revenue of ₹459,804,176.78. Electronics is the largest revenue category (49.27% of revenue) but has the steepest January-to-June decline of 35.0%. Apparel has the strongest growth at 34.1%, while Grocery is the most consistent by monthly coefficient of variation.

At product level, Smart TV is the lowest-revenue Electronics product at ₹41,882,166.21. In Apparel, Saree and T-Shirt are the top two products by six-month revenue. Discount analysis shows average revenue per transaction falling 22.5% from 0% to 20% discount, while average units sold remain nearly flat. This supports scrutiny of deeper discounts, but profitability cannot be measured without cost or margin data.

2. Business Context and Assignment Alignment

The source brief describes a regional retail chain with 12 stores across Andhra Pradesh and five categories. The Category Manager is preparing for a quarterly supplier review and needs evidence on category revenue trends, top/bottom products, discount effectiveness, day-of-week/month patterns, and one clear category recommendation.

The notebook follows the requested sequence and adds the requested ETL cleaning stage before EDA. Each required section contains runnable code plus plain-English business interpretation.

3. CBIM Problem Canvas

Canvas Element	Answer
Situation (S)	A 12-store retail chain is preparing for a quarterly supplier review across five categories; the manager needs evidence for shelf-space and promotion decisions.
Complication (C)	The manager needs to distinguish growing, stable and declining categories and identify product-level drivers and weak spots.
Question (Q)	Which categories are growing/declining, which products drive/drag performance, whether discounts are associated with higher transaction revenue, and when promotions should be timed?
SSOT	The supplied MUC01 retail sales transaction dataset.
Success definition	A clear category priority, specific challenge supported by revenue, discount finding backed by numbers, timing signal, and limitation.
Primary stakeholder	The Category Manager preparing for the supplier review.

4. ETL and Data Cleaning

The analysis begins with Extract → Transform → Load → Analyze. The raw CSV is extracted into Pandas; text and numeric fields are standardised; dates are parsed; business rules are validated; analytical features are created; the clean dataframe is used as the analysis table; and a cleaned CSV is exported for reproducibility.

Check	Result
Rows loaded	107,836

Rows after cleaning	107,836
Duplicate transaction IDs	0
Rows with at least one null	0
Critical validation issues	0
Revenue mismatches > ₹0.01	0
Unique products	25
Unique categories	5
Unique stores	12
Unique cities	6

Revenue was reconciled against $\text{units_sold} \times \text{unit_price} \times (1 - \text{discount_pct}/100)$, using ₹0.01 tolerance for stored rounding. No material mismatches were found. No substantive rows required removal, so the cleaning stage is primarily validation and feature transformation.

5. Dataset Profile

Metric	Value
Transactions	107,836
Total Revenue	₹459,804,176.78
Total Units Sold	278,593
Date Range	01 Jan 2026 – 30 Jun 2026
Products	25
Categories	5
Stores	12
Cities	6

6. Category Revenue Trend

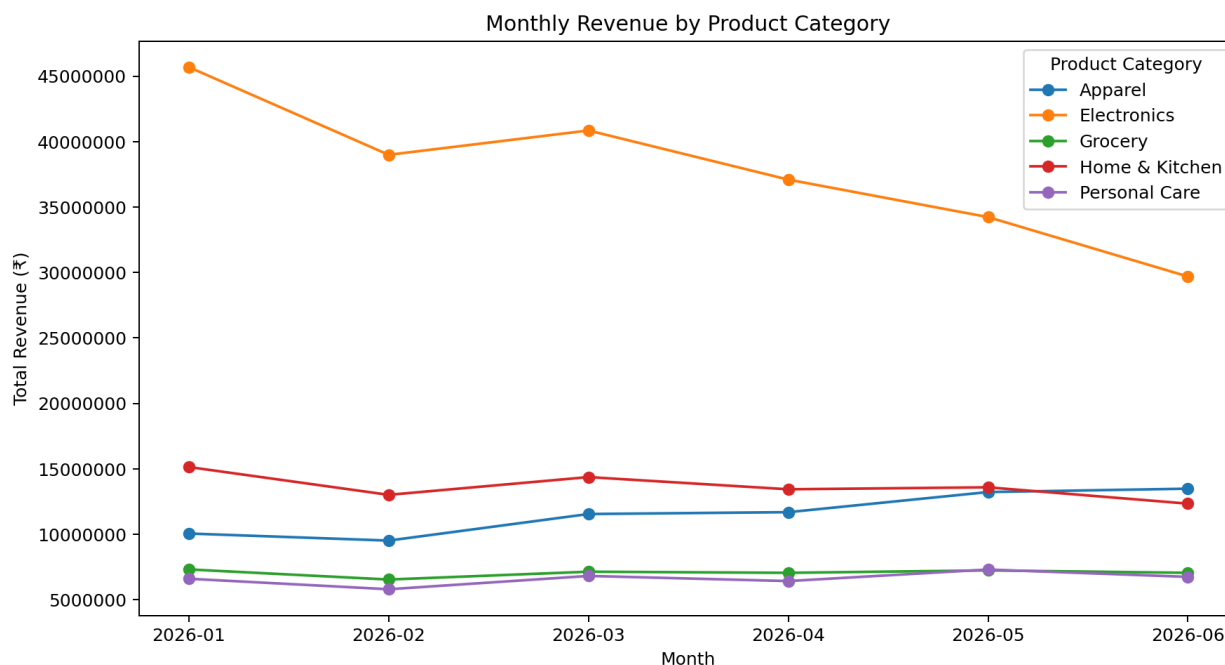


Figure 1. Monthly revenue by product category.

Category	January Revenue (₹)	June Revenue (₹)	Change (₹)	Change (%)	Monthly CV (%)
Electronics	₹45,686,596.63	₹29,686,824.81	₹-15,999,771.82	-35.02%	14.60%
Home & Kitchen	₹15,130,871.54	₹12,333,283.07	₹-2,797,588.47	-18.49%	7.26%
Grocery	₹7,307,011.34	₹7,047,719.23	₹-259,292.11	-3.55%	3.87%
Personal Care	₹6,593,734.81	₹6,737,354.12	₹143,619.31	2.18%	7.52%

Apparel	₹10,049,916.78	₹13,476,952.88	₹3,427,036.10	34.10%	13.88%
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Electronics declined from ₹45,686,596.63 in January to ₹29,686,824.81 in June, a 35.0% decrease. Apparel increased 34.1%, the strongest positive change. Grocery has the lowest monthly CV at 3.87%, indicating the most consistent monthly revenue.

7. Top and Bottom Products by Category

Category	Top 1 Product	Top 1 Revenue (₹)	Top 2 Product	Top 2 Revenue (₹)	Bottom 1 Product	Bottom 1 Revenue (₹)	Bottom 2 Product	Bottom 2 Revenue (₹)
Apparel	Saree	₹14,161,864.63	T-Shirt	₹14,099,251.65	Kurta	₹13,606,747.03	Formal Shirt	₹13,748,976.91
Electronics	Headphones	₹47,696,053.99	Laptop	₹46,551,078.39	Smart TV	₹41,882,166.21	Tablet	₹44,744,388.93
Grocery	Rice (5kg)	₹8,562,171.86	Dal (1kg)	₹8,488,303.64	Sugar (1kg)	₹8,377,786.05	Oil (1L)	₹8,431,630.01
Home & Kitchen	Mixer Grinder	₹16,903,254.58	Storage Box	₹16,676,535.31	Water Bottle	₹15,825,842.17	Pressure Cooker	₹16,142,329.39
Personal Care	Shampoo	₹8,168,328.26	Body Lotion	₹8,106,578.00	Sunscreen	₹7,593,863.45	Toothpaste	₹7,768,812.48

Electronics is the category to challenge in the supplier review because its category trend is the weakest and Smart TV is the lowest-revenue product within it at ₹41,882,166.21. Apparel's top products, Saree (₹14,161,864.63) and T-Shirt (₹14,099,251.65), provide product-level support for its growth.

8. Discount Effectiveness

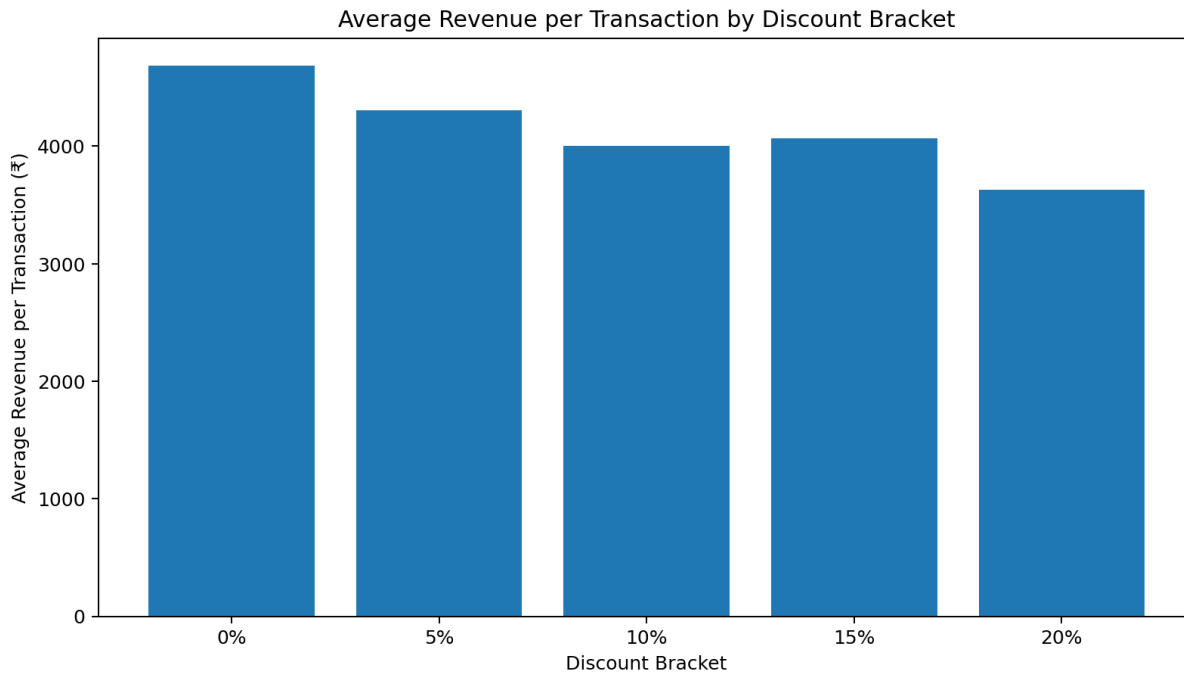


Figure 2. Average revenue per transaction by discount bracket.

Discount Bracket	Transactions	Avg. Revenue per Transaction	Avg. Units Sold
0%	36,033	₹4,685.65	2.59
5%	23,922	₹4,303.73	2.59
10%	23,810	₹4,005.55	2.57
15%	11,935	₹4,069.16	2.60
20%	12,136	₹3,631.73	2.58
25%+	0	—	—

Average revenue per transaction decreases from ₹4,685.65 at 0% discount to ₹3,631.73 at 20%, a 22.5% reduction.

Average units sold change only from 2.59 to 2.58. Thus, higher discount brackets do not show proportionate transaction uplift. This is an association, not a causal estimate, and the 25%+ bracket has no observations.

The dataset has no product cost or gross-margin field, so actual margin erosion cannot be quantified. Product mix, store mix and promotion targeting can also influence the observed averages.

9. Day-of-Week and Monthly Patterns

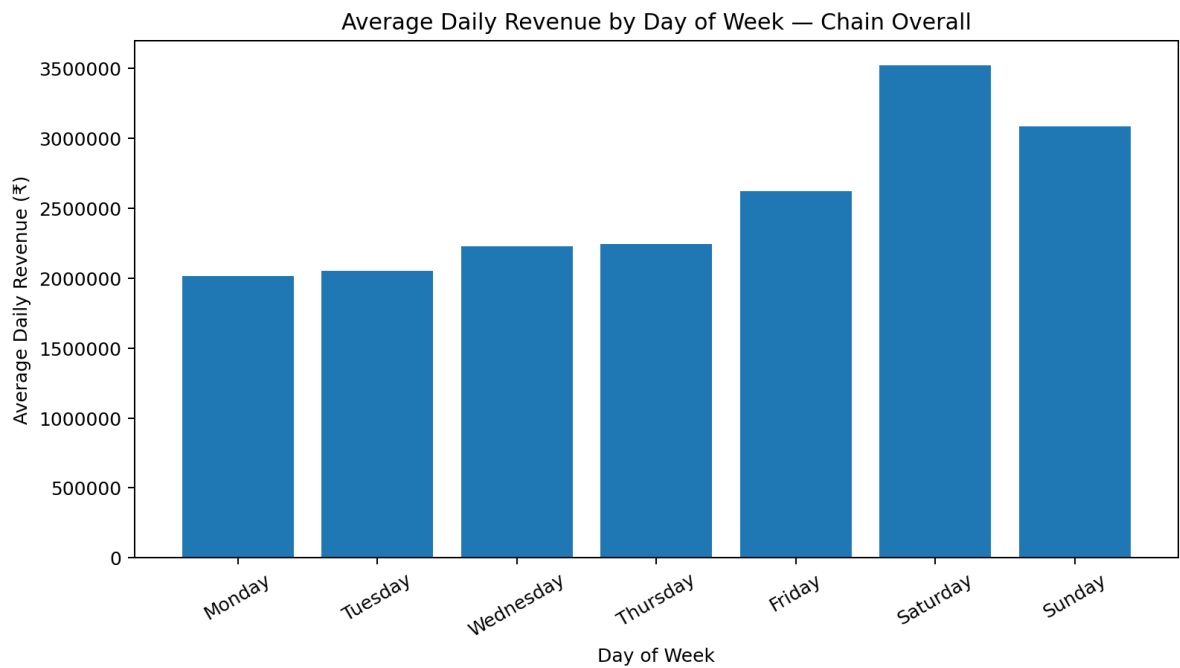


Figure 3. Average daily revenue by weekday — chain overall.

Day of Week	Average Daily Revenue (₹)
Monday	₹2,012,140.96
Tuesday	₹2,054,129.70
Wednesday	₹2,228,117.24
Thursday	₹2,245,493.02
Friday	₹2,623,611.89
Saturday	₹3,523,605.44
Sunday	₹3,083,374.58

Saturday is the strongest weekday at ₹3,523,605.44 average daily revenue, while Tuesday is the weakest at ₹2,012,140.96.

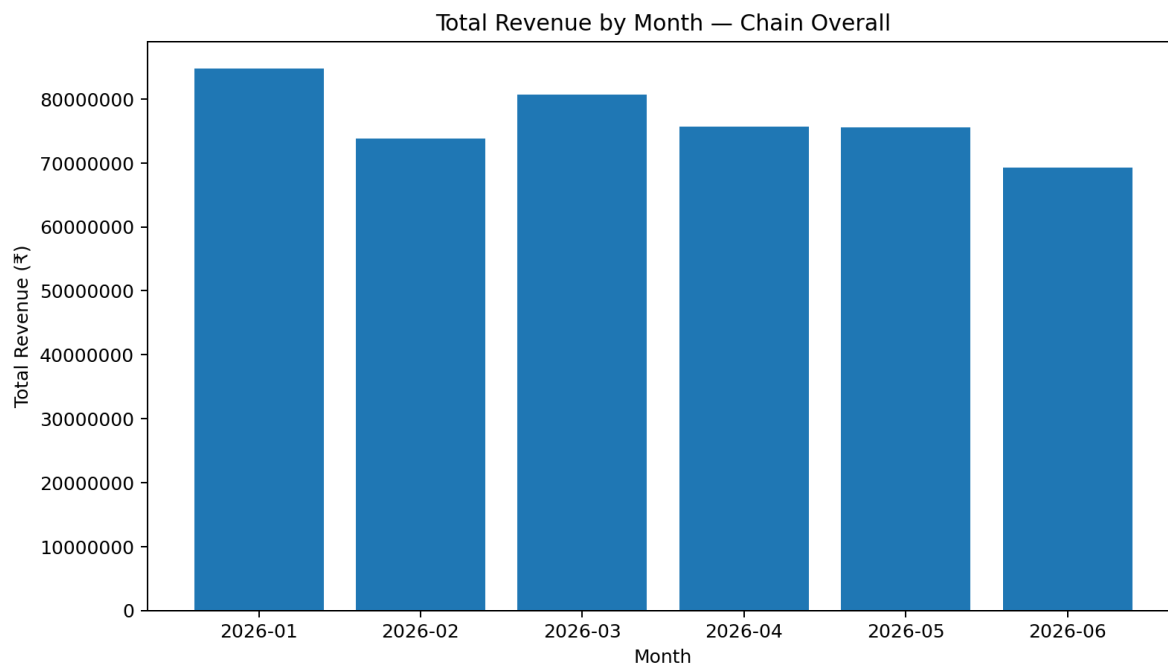


Figure 4. Total chain revenue by month.

January is the strongest month at ₹84,768,131.10; June is the weakest at ₹69,282,134.11.

10. Electronics Timing Pattern

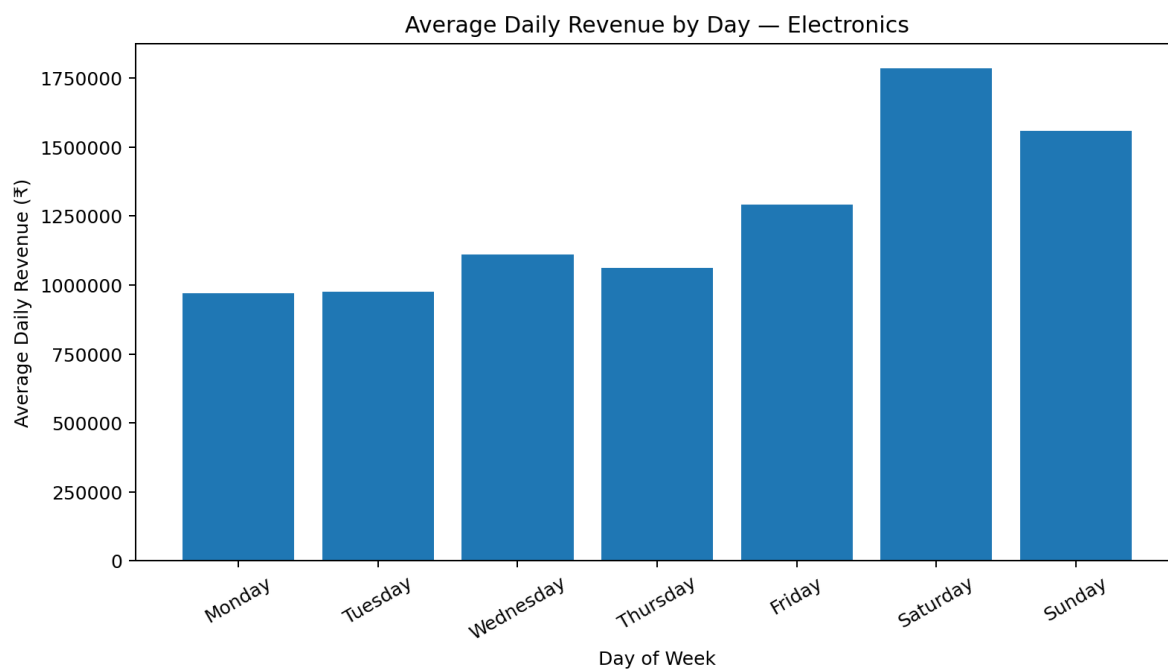


Figure 5. Average daily revenue by weekday — Electronics.

Electronics follows the chain's weekday pattern: Saturday is strongest at ₹1,786,658.45 average daily revenue and Tuesday is weakest at ₹970,975.52. Its monthly total peaks in January at ₹45,686,596.63 and bottoms in June at

₹29,686,824.81. This alignment supports weekend-focused promotional timing, with targeted experiments on weaker days.

11. Category Manager Briefing

- Apparel has the strongest growth case for controlled incremental investment testing because revenue rose 34.1% from January to June, supported by products such as Saree and T-Shirt.
- Electronics should be challenged in the supplier review: revenue declined 35.0% and Smart TV generated ₹41,882,166.21 over six months.
- Discount discussions should focus on the lack of proportionate transaction uplift: average revenue per transaction is 22.5% lower at 20% discount than at 0%, while average units sold are nearly flat.
- Promotional timing should favour Saturdays, the strongest average daily revenue day for both the chain and Electronics.
- Before a final investment decision, obtain product cost/gross-margin, inventory/stock-out and promotion-exposure data so profitability and incremental lift can be assessed.

12. Action Plan

Area	Action
Investment test	Prioritise Apparel for controlled incremental shelf/promotion testing.
Supplier challenge	Challenge Electronics on the January-to-June decline and Smart TV contribution.
Discount review	Require evidence of incremental demand before supporting deeper discounts.
Timing	Emphasise Saturdays; use weaker days for targeted experiments.
Additional data	Request cost/margin, inventory/stock-out and promotion-exposure data.

13. Limitations

- The dataset has no product cost/gross margin, so profitability cannot be calculated.
- Discount brackets are observational; product mix, store mix and promotion targeting can affect averages.
- The 25%+ discount bracket has no observations.
- Only January–June 2026 is covered, so longer-term seasonality cannot be assessed.
- No inventory, stock-out, campaign-exposure or customer-level variables are available.

14. Deliverables

This package includes the end-to-end notebook, a submission-ready one-page insight summary, a detailed report, and the cleaned CSV supporting the ETL stage.